ANVIL COMPOUNDS

RESEARCH USE ONLY

● RESEARCH GRADE · 99%+ PURITY · USA-BASED

RESEARCH REFERENCE & SAFETY DOCUMENTATION

Tesamorelin

Stabilized GHRH(1–44) analogue · Research-grade reference material

CLASS	LENGTH	MODIFICATION	MW
GHRH analogue	**44 amino acids**	**trans-3-hexenoyl, N-terminal**	**≈5136 g/mol**

01 COMPOUND AT A GLANCE

Identity & Composition

Tesamorelin is a synthetic analogue of the full 44-amino-acid human growth hormone-releasing hormone, GHRH(1–44), bearing a trans-3-hexenoic acid group conjugated to the N-terminus. This modification confers resistance to dipeptidyl peptidase-IV (DPP-IV) cleavage, extending stability while preserving activity at the GHRH receptor.

Product name	Tesamorelin
Class	Growth hormone-releasing factor (GHRH) analogue
Parent peptide	Human GHRH(1–44)
Modification	trans-3-hexenoic acid at N-terminal tyrosine
Length	44 amino acids
Molecular weight	≈5136 g/mol
PubChem CID	16137828
Appearance	White lyophilized powder
Identified use	In-vitro research & development only

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YADAIFTNSYRKVLGQLSARKLLQDIMSRQQGESN
QERGARARL
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GHRH(1–44) primary sequence. Tesamorelin corresponds to this sequence with a trans-3-hexenoyl group at the N-terminus. Confirm the exact modification and salt form of the material supplied against the lot documentation.

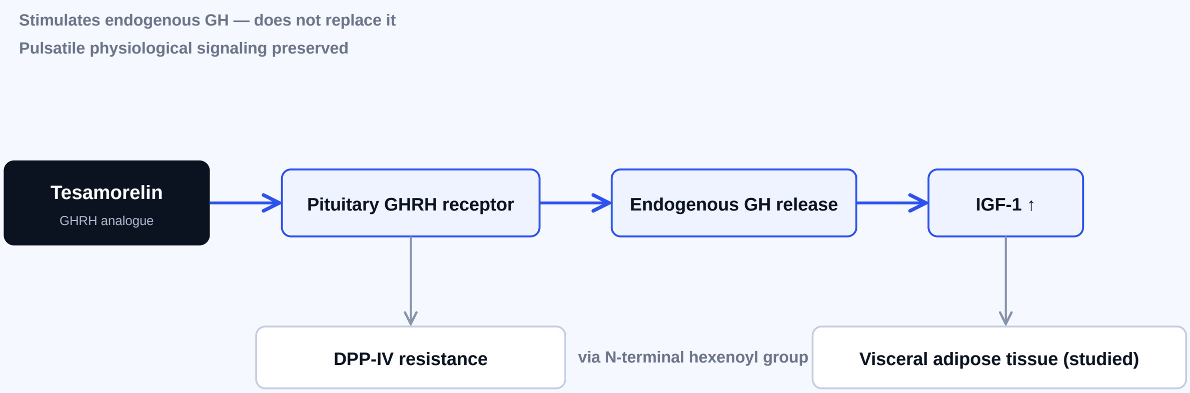
> **Important distinction.** Research-grade tesamorelin supplied as a reference material is **not** the FDA-approved drug product. The approved product is a specific manufactured formulation subject to pharmaceutical quality standards; research-grade material is supplied for laboratory use only and is not equivalent to, or a substitute for, an approved medicine.

Research Background

Developed by Theratechnologies (Montreal), tesamorelin is mechanistically distinct from exogenous growth hormone: it acts upstream at the pituitary GHRH receptor to stimulate the body's own GH secretion, preserving pulsatile physiological signaling rather than replacing the hormone directly.

Mechanism

- **GHRH receptor agonism.** Binds GHRH receptors on the anterior pituitary with high affinity, stimulating endogenous growth hormone secretion.
- **Downstream IGF-1.** Increased GH raises insulin-like growth factor 1 (IGF-1).
- **DPP-IV resistance.** The N-terminal trans-3-hexenoic acid modification protects against dipeptidyl peptidase-IV cleavage, extending half-life relative to unmodified GHRH (reported ≈26 min vs ≈12 min for sermorelin).
- **Adipose effects studied.** In clinical trials the studied outcome was reduction of visceral adipose tissue, distinct from general weight loss.
- **Requires intact pituitary axis.** Because it acts upstream, activity depends on a functional hypothalamic-pituitary response.



Established mechanism of action for the tesamorelin molecule. Schematic. Research-grade material supplied here is for in-vitro laboratory use only.

Comparison note. Tesamorelin is the full GHRH(1–44) with a stability modification; sermorelin is the shorter GHRH(1–29) fragment. The two are not interchangeable in research design.

What the Literature Supports — and What It Doesn’t

Tesamorelin is unusual in this catalogue: unlike most research peptides, the molecule has completed pivotal human trials and holds FDA approval. That approval is real — and it is also **narrow**. Both halves of that statement matter.

Evidence at a glance

PHASE III HUMAN DATA	FDA STATUS	APPROVED INDICATION	WEIGHT-LOSS INDICATION
Available (n≈806)	**Approved (2010)**	**HIV lipodystrophy only**	**Explicitly none**

- **Pivotal trials.** In two Phase III trials (Falutz et al.), tesamorelin 2 mg daily reduced visceral adipose tissue by approximately 15% versus placebo over 26 weeks (pooled treatment effect −15.4%, $p<0.001$, n=806), with accompanying lipid changes.
- **The approved indication is narrow:** reduction of excess abdominal fat in adults with HIV-associated lipodystrophy. Approved 2010 (Egrifta), reformulated 2019 (Egrifta SV), with an F8 formulation approved 2025 (Egrifta WR).
- **The label states it is not indicated for weight-loss management** and is weight neutral. Any framing of tesamorelin as a weight-loss agent contradicts its own prescribing information.
- **Long-term cardiovascular safety has not been established** per the prescribing information. Contraindications include active malignancy, pituitary disruption, and pregnancy.
- Use outside HIV lipodystrophy is **off-label with thinner evidence**; research in non-HIV visceral and hepatic fat contexts is ongoing.

> **Appropriate characterization:** a well-characterized GHRH analogue with genuine Phase III human evidence supporting one specific approved indication. Research-grade material supplied here is not the approved drug product and is intended solely for in-vitro laboratory research.

Regulatory & status notes

US FDA	Approved (Egrifta 2010; Egrifta SV 2019; Egrifta WR / F8 2025) for reduction of excess abdominal fat in adults with HIV-associated lipodystrophy. Prescription medicine.
Label limitation	Not indicated for weight-loss management; weight neutral. Long-term cardiovascular safety not established.
Supply basis	Research reagent for in-vitro laboratory use only; not the approved drug product; not for human or veterinary administration.

Selected references

1. Falutz J, et al. Pivotal Phase III trials of tesamorelin in HIV-associated lipodystrophy (pooled analysis, n=806).
2. EGRIFTA SV (tesamorelin for injection) US Prescribing Information. FDA/DailyMed.
3. PubChem compound record, tesamorelin (CID 16137828).
4. FDA approval of the F8 formulation (EGRIFTA WR), 2025.

Safety Data Sheet

Prepared per GHS Rev. 8 / OSHA HazCom 2012
(29 CFR 1910.1200) · 16-section format

Note. This Safety Data Sheet is compiled from public sources and weight-of-evidence assessment. Independent review by a qualified chemical-safety professional is recommended prior to use. Supplier identification and emergency-contact details must be completed by the supplier.

1 Identification

Product	Tesamorelin
Class	GHRH(1–44) analogue, N-terminal trans-3-hexenoyl
MW	≈5136 g/mol
PubChem CID	16137828
Use	In-vitro research only
Restriction	Not for human/veterinary, food, drug, cosmetic, clinical, or therapeutic use. Research-grade material is not the approved drug product.
Supplier	
Emergency	

2 Hazard Identification

Not classified based on currently available data for the research-grade substance. **Signal word:** None. **Pictograms:** None required. **Precautionary:** P261, P264, P280, P501. Handle as a potentially bioactive peptide; note that the molecule is pharmacologically active at the GHRH receptor and should be handled accordingly.

3 Composition

Single-substance product, >98% research grade. Residual synthesis reagents, solvents and counter-ions may be present consistent with research-grade material. See lot COA for impurity profile.

4 First-Aid Measures

Eyes: rinse with water. **Skin:** wash with soap and water; remove contaminated clothing. **Inhalation:** fresh air. **Ingestion:** rinse mouth; do not induce vomiting unless directed. Treat symptomatically; no specific antidote known.

5 Fire-Fighting

Media: CO_2, dry chemical, foam, water spray. Combustion may produce CO, CO_2, NOx and sulfur oxides. Use SCBA and full protective gear.

6 Accidental Release

Avoid dust; ventilate; use PPE per §8. Prevent entry to drains/waterways. Collect in a closed labeled container for disposal per §13.

7 Handling & Storage

Handle as a pharmacologically active peptide; minimize all routes of exposure. Weigh/reconstitute in fume hood. Store lyophilized solid tightly closed, protected from light/moisture/heat; −20 °C long-term, 2–8 °C short-term. Aliquot reconstituted solutions; store frozen, protected from light. Observe COA retest date.

8 Exposure Controls / PPE

No established OEL; control to ALARA. Because the substance is pharmacologically active at low quantities, containment when handling dry powder is particularly important. Fume hood or ventilated balance enclosure; nitrile gloves (double-glove for neat powder); ANSI Z87.1 eye protection; lab coat; N100/P100 respirator where controls insufficient.

9 Physical & Chemical Properties

State	Lyophilized powder
Appearance	White powder
Odor	Odorless
Solubility	Soluble in water
MW	≈5136 g/mol
MP/BP/Flash	Not determined

10 Stability & Reactivity

Stable under normal storage. Avoid heat, moisture, sunlight, UV, open flame. Incompatible with strong oxidizers, strong acids/bases, strong reducing agents. Methionine-containing sequence is susceptible to oxidation. Decomposition: CO, CO_2, NOx, SOx. No hazardous polymerization.

11 Toxicological Information

Human safety data exists for the approved drug product under clinical supervision and is documented in its prescribing information; that data does not characterize occupational exposure to the research-grade substance. No occupational LD_{50}/LC_{50} available. Skin/eye: treat as potentially irritating. Mutagenicity, carcinogenicity, reproductive toxicity, STOT, aspiration: not classified based on available data; hazards cannot be excluded. Avoid exposure if pregnant, nursing, or planning pregnancy.

12 Ecological Information

No substance-specific ecotoxicity, persistence or bioaccumulation data identified. Avoid release to surface water, soil, drains or sewers.

13 Disposal

Dispose per applicable local/state/federal regulations; US per 40 CFR 261–270 (RCRA). Do not dispose into sewers/waterways. Use a licensed waste disposal company.

14 Transport Information

Not regulated as dangerous goods under DOT (49 CFR), IATA or IMDG based on current classification. UN number: not applicable. Shipper must independently verify.

15 Regulatory Information

The tesamorelin *molecule* is an FDA-approved prescription drug substance (see Section 03). Material supplied here is research grade and is not an approved drug product, is not manufactured to pharmaceutical standards, and must not be represented or used as a medicine. US TSCA: may qualify for R&D exemption (40 CFR 720.36). OSHA HazCom 2012 aligned with GHS Rev. 8. EU REACH: supplied for SR&D use; verify exemption.

16 Other Information

Revision [DATE] · Version 1.0. Independent professional review recommended prior to use. Sources: PubChem; FDA prescribing information; peer-reviewed literature; OSHA HazCom 2012; GHS Rev. 8. Provided without warranty. For scientific R&D use only.